

Identify / Predict / Derive Trichotomy Position Document v0.1

Author: Keeper Filed: 2026-05-22 Friday 08:25 EDT (FAST CADENCE, Wednesday substrate review owed item)

The trichotomy

BST claims fall into three honest epistemic categories distinguished by mechanism state:

IDENTIFY (substrate-natural match)

BST identifies a numerical value as substrate-natural — i.e., expressible in BST primary integers — but the *mechanism* by which the substrate produces that value is not yet explicit.

Example: 21cm wavelength = $N_{c\cdot g}$ cm (Elie 2026-04). Numerical match exists; mechanism is “the substrate hydrogen hyperfine transition expresses $N_{c\cdot g}$ via [TBD].”

Tier: I-tier (identification).

PREDICT (mechanism present, value derived but unverified)

BST predicts a numerical value via explicit substrate mechanism, but the experimental measurement either disagrees or hasn’t been made yet.

Example: $n_s = 1 - n_C/N_{\max} = 0.9635$ (Elie Thursday). Mechanism explicit (CMB spectral tilt = primary-integer correction); Planck 2018 measures 0.9649; 0.32σ deviation = within experimental error but technically predicts a specific value.

Tier: D-tier-prediction (derived prediction) or C-tier (conditional, depends on future measurement).

DERIVE (mechanism present, value matches measurement EXACT or at $<1\%$)

BST derives a numerical value via explicit substrate mechanism AND the value matches experimental measurement EXACT or at sub-percent precision.

Example: $m_p/m_e = 6\pi^5 = 938.272$ MeV at 0.002% (T2438). Mechanism: Bergman bulk/fiber volume ratio per K88 + 4-Zone vacuum + Casimir spectrum. Match: 0.002%.

Tier: D-tier (derived).

Why the trichotomy matters

External presentation discipline (Cal #50 + Casey curvature principle): different tiers carry different epistemic weights for venue submission.

- D-tier derives are the publishable core
- I-tier identifications require mechanism completion before promotion
- C-tier predictions require experimental falsification (or confirmation) before D-promotion

Mixing tiers in venue submission risks: - I-tier overcounted as derivation (cherry-picking concern) - D-tier diluted by I-tier (rigor concern) - C-tier promoted prematurely (predictive overreach concern)

The trichotomy keeps the methodology stack honest about what’s actually demonstrated vs identified vs predicted.

Current catalog distribution (Friday 08:25 EDT)

Per Grace Friday morning: catalog at 4738 invariants, 78% D-tier (honest).

Approximate trichotomy distribution: - D-tier (derive): ~3,696 entries (78%) - I-tier (identify): ~700 entries (~15%)
- C-tier (predict) + S-tier (structural): ~342 entries (~7%)

D-tier 78% is the publishable core.

Operational rules

Per Cal Appendix G + standing methodology:

1. D-tier promotion requires mechanism: I-tier $\rightarrow$ D-tier only after substrate-mechanism derivation explicit
 2. C-tier requires falsifier: C-tier predictions need explicit experimental falsifier path (e.g., n_s requires next-gen CMB)
 3. No mixing in venue submission: papers distinguish identification vs derivation vs prediction
 4. Tier-label appendix in all papers: Appendix G discipline (Casey approved Wednesday May 16 evening)
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Position for Friday work

All Friday flagship outputs should be tier-labeled at production: - Lyra T2452 sub-substrate hierarchy: D-tier (mechanism = Mersenne identity + uniqueness verification) - Lyra T2453 cross-Cartan α -analog universal formula: D-tier (formal theorem, derivation explicit) - Lyra T2454 churn hole universal: D-tier (T2439 generalization) - Lyra T2455 c_{FK} universal: D-tier (T2442 generalization) - Grace Graph Forces empirical ($p \approx 2.7 \times 10^{-5}$): D-tier-empirical (statistical mechanism + hypergeometric null) - Elie $n_s = 0.9635$: C-tier-prediction (mechanism present + experimental match within error)

All output documented with explicit tier in Operational Spec discipline.

— Keeper, Identify/Predict/Derive position v0.1 filed Friday 08:25 EDT (FAST CADENCE)